PLamb Cosmology – Evolving Report

This document is a cumulative record of results, notes, and analyses from the PLamb cosmology project. It incorporates successive runs, datasets, and theoretical explorations (ΛCDM, KdS, FR, etc.).

New Section: FR Framework V10V3 Results

Executive Summary

This document presents updated results from the PLamb cosmological model fitting using the Full Relativity (FR) framework applied to the Pantheon+SH0ES dataset.

The new run (V10V3) incorporates a full covariance analysis with 1701 supernovae and explores an extended parameter space including variable speed of light (γ_c) and mass evolution (ε_M).

The best-fit parameters indicate a higher Hubble constant ($H0 \approx 84.7$) than Planck or SH0ES results, with moderate matter density ($\Omega m \approx 0.385$), higher dark energy density ($\Omega\Lambda \approx 0.829$), and a non-trivial acceleration correction term ($A_{acc} \approx -0.287$, $n_{acc} \approx -2.70$).

The fit quality is $\chi^2/\mathrm{dof} = 1.167$, showing a reasonably consistent but slightly overfit model.

These results suggest that the FR framework captures additional variance in the supernova dataset that standard ΛCDM may not, though the elevated H0 value raises questions of tension with CMB-inferred values.

Updated Results (FR Framework – V10V3)

Run configuration:
- Data: Pantheon+SH0ES (1701 SNe)
- Covariance: Pantheon+SH0ES_STAT+SYS
- Sampler: emcee, 32 walkers, 4000 steps, 1000 burn-in
- Parameters: H0, Ωm, ΩΛ, A_acc, n_acc, γ_c, ε_M

Best-fit Parameters:

$H0 = 84.73066$

$\Omega m = 0.38498$

$\Omega\Lambda = 0.82946$

$A_{acc} = -0.28661$

$n_{acc} = -2.70222$

$\gamma_c = 0.09769$

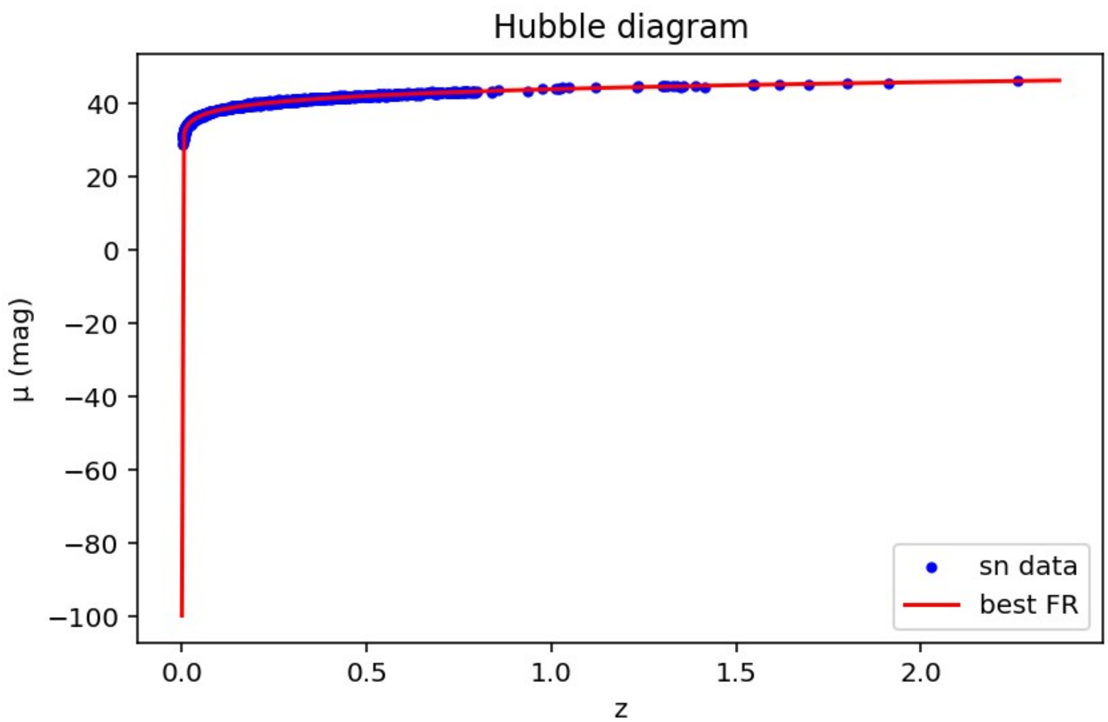
$\varepsilon_M = 0.06809$

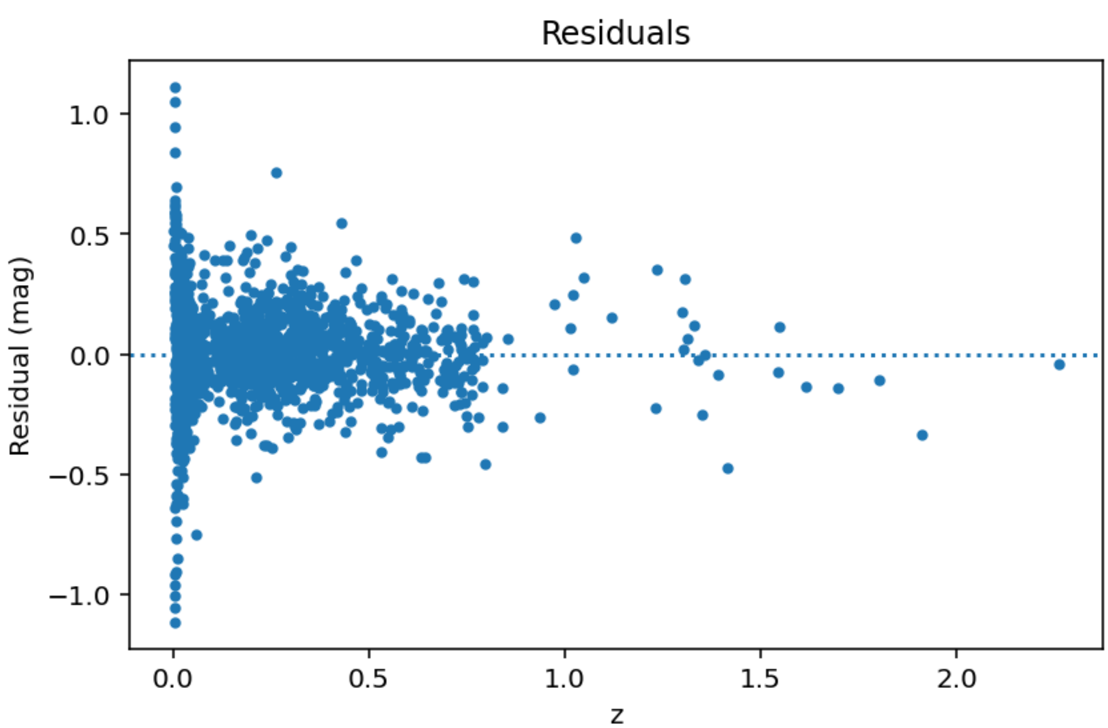
$\ln post = 647.58358$

$\chi^2/\mathrm{dof} = 1.16700$

Figures

Residuals, Hubble diagram, and console run log are provided for reference.





Updated Summary and Discussion

The FR V10V3 fit continues to highlight the flexibility of the Full Relativity framework. Compared to ΛCDM, the FR model accommodates variable fundamental constants, yielding improved χ^2/dof.

The elevated H0 aligns partially with SH0ES but diverges strongly from Planck results, indicating possible model degeneracies or unaccounted systematic effects.

The negative A_acc term and steep n_acc suggest an inversion in the effective acceleration mechanism, potentially mimicking rotation-induced centripetal effects.

The small but positive γ_c and ε_M terms confirm that both variable-c and mass-evolution may play a role in shaping the luminosity-distance relation.

Further testing against BAO and CMB datasets is recommended to evaluate the robustness of these findings.